JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON WEDNESDAY 28th JANUARY 2026) **TIME : 3:00 PM TO 6:00 PM**

MATHEMATICS — TEST PAPER WITH ANSWER

SECTION-A

1. Given below two statements :

Statement I : $25^{13} + 20^{13} + 8^{13} + 3^{13}$ is divisible by 7.

Statement II : The integral part of $\left(7+4\sqrt{3}\right)^{25}$ is an odd number.

In the light of the above statements, choose the **correct answer** from the options given below :

(1) Both **Statement I** and **Statement II** are false.

(2) Both **Statement I** and **Statement II** are true.

(3) **Statement I** is false but **Statement II** is true.

(4) **Statement I** is true but **Statement II** is false.

2. The sum of the coefficients of x^{499} and x^{500} in $(1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \ldots\ldots + x^{1000}$ is

(1) ${}^{1001}C_{501}$ (2) ${}^{1002}C_{500}$

(3) ${}^{1002}C_{501}$ (4) ${}^{1000}C_{501}$

3. Let A be the focus of the parabola $y^2 = 8x$. Let the line $y = mx + c$ intersect the parabola at two distinct points B and C. If the centroid of the triangle ABC is $\left(\frac{7}{3}, \frac{4}{3}\right)$, then $(BC)^2$ is equal to :

(1) 41 (2) 80

(3) 89 (4) 32

4. The probability distribution of a random variable X is given below :

X	4k	$\frac{30}{7}$k	$\frac{32}{7}$k	$\frac{34}{7}$k	$\frac{36}{7}$k	$\frac{38}{7}$k	$\frac{40}{7}$k	6k
P(X)	$\frac{2}{15}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{5}$	$\frac{1}{15}$

If $E(X) = \frac{263}{15}$, then $P(X < 20)$ is equal to :

(1) $\frac{3}{5}$ (2) $\frac{8}{15}$

(3) $\frac{11}{15}$ (4) $\frac{14}{15}$

5. Let $f(x) = \lim\limits_{\theta\to 0}\left(\frac{\cos\pi x - x^{\left(\frac{2}{\theta}\right)}\sin(x-1)}{1+x^{\left(\frac{2}{\theta}\right)}(x-1)}\right), x \in R$.

Consider the following two statements :

(I) f(x) is discontinous at x = 1.

(II) f(x) is continous at x = - 1.

Then,

(1) Neither (I) nor (II) is True

(2) Both (I) and (II) are True

(3) Only (II) is True

(4) Only (I) is True

6. Considering the principal values of inverse trigonometric functions, the value of the expression $\tan\left(2\sin^{-1}\left(\frac{2}{\sqrt{13}}\right) - 2\cos^{-1}\left(\frac{3}{\sqrt{10}}\right)\right)$ is equal to :

(1) $-\frac{33}{56}$

(2) $\frac{33}{56}$

(3) $\frac{16}{63}$

(4) $-\frac{16}{63}$

7. Let the arithmetic mean of $\frac{1}{a}$ and $\frac{1}{b}$ be $\frac{5}{16}$, $a > 2$.

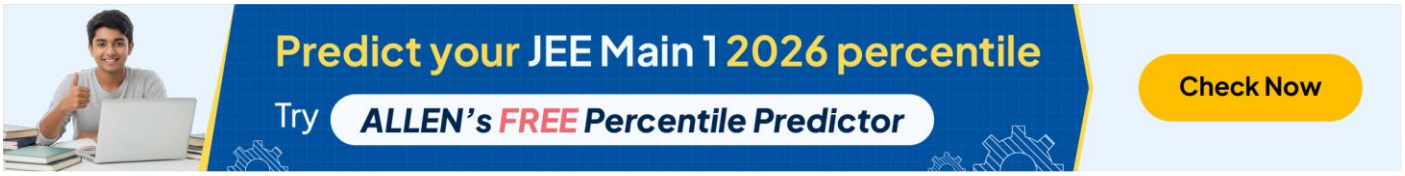
If α is such that a, 4, α, b are in A.P., then the equation $\alpha x^2 - ax + 2(\alpha - 2b) = 0$ has :

(1) One root in (1,4) and another in (–2,0)

(2) One root in (0,2) and another in (– 4, –2)

(3) Complex roots of magnitude less than 2

(4) Both roots in the interval (–2, 0)



8. Given below are two statements :

Statement I : The function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) = \frac{x}{1+|x|}$ is one-one.

Statement II : The function $f : \mathbf{R} \to \mathbf{R}$ defined by $f(x) = \frac{x^2+4x-30}{x^2-8x+18}$ is many-one.

In the light of the above statements, choose the **correct answer** from the options given below :

(1) Both **Statement I** and **Statement II** are false.

(2) Both **Statement I** and **Statement II** are true.

(3) **Statement I** is false but **Statement II** is true .

(4) **Statement I** is true but **Statement II** is false.

9. An ellipse has its center at (1,-2), one focus at (3,-2) and one vertex at (5, - 2). Then the length of its latus rectum is :

(1) $\frac{16}{\sqrt{3}}$ (2) 6

(3) $4\sqrt{3}$ (4) $6\sqrt{3}$

10. Let the ellipse $E: \frac{x^2}{144}+\frac{y^2}{169}=1$ and the hyperbola $H : \frac{x^2}{16}-\frac{y^2}{\lambda^2}=-1$ have the same foci. If e and L respectively denote the eccentricity and the length of the latus rectum of H, then the value of 24(e + L) is :

(1) 296 (2) 126

(3) 148 (4) 67

11. Let $P_1 : y = 4x^2$ and $P_2 : y = x^2 + 27$ be two parabolas. If the area of the bounded region enclosed between P_1 and P_2 is six times the area of the bounded region enclosed between the line $y = \alpha x$, $\alpha > 0$ and P_1, then α is equal to :

(1) 8 (2) 15

(3) 12 (4) 6

12. Let the circle $x^2 + y^2 = 4$ intersect x-axis at the points A(a, 0), a > 0 and B(b,0). Let P(2 cosα, 2 sinα), $0 < \alpha < \frac{\pi}{2}$ and Q(2 cosβ, 2 sinβ) be two points such that $(\alpha - \beta) = \frac{\pi}{2}$. Then the point of intersection of AQ and BP lies on :

(1) $x^2 + y^2 - 4y - 4 = 0$

(2) $x^2 + y^2 - 4x - 4 = 0$

(3) $x^2 + y^2 - 4x - 4y = 0$

(4) $x^2 + y^2 - 4x - 4y - 4 = 0$

13. Let [·] denote the greatest integer function. Then $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{12(3+[x])}{3+[\sin x]+[\cos x]}\right) dx$ is equal to:

(1) $15\pi + 4$ (2) $11\pi + 2$

(3) $13\pi + 1$ (4) $12\pi + 5$

14. Let y = y(x) be the solution of the differential equation $x\frac{dy}{dx} - y = x^2 \cot x, x \in (0, \pi)$.

If $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$, then $6y\left(\frac{\pi}{6}\right) - 8y\left(\frac{\pi}{4}\right)$ is equal to :

(1) 3π (2) -3π

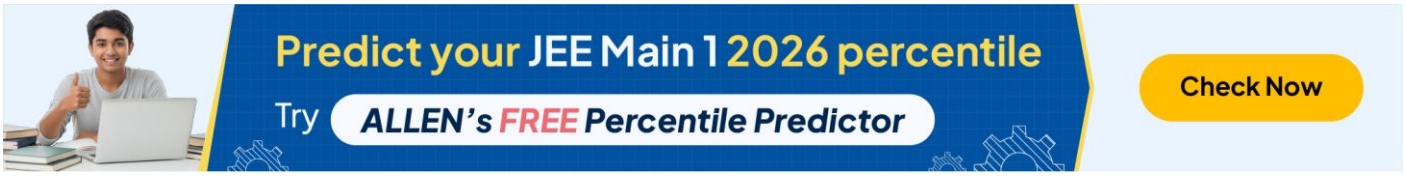
(3) $-\pi$ (4) π

15. The sum of all the elements in the range of f(x)=Sgn(sinx)+Sgn(cosx)+Sgn(tanx)+Sgn(cotx), $x \neq \frac{n\pi}{2}$, $n \in \mathbf{Z}$,

where $\text{Sgn}(t) = \begin{cases} 1, & \text{if } t > 0 \\ -1 & \text{if } t < 0 \end{cases}$, is

(1) 4 (2) 2

(3) –2 (4) 0



16. Let Q(a, b, c) be the image of the point P(3, 2, 1) in the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$. Then the distance of Q from the line $\frac{x-9}{3}=\frac{y-9}{2}=\frac{z-5}{-2}$ is

(1) 6 (2) 8

(3) 7 (4) 5

17. Let P be a point in the plane of the vector $\overrightarrow{AB}=3\hat{i}+\hat{j}-\hat{k}$ and $\overrightarrow{AC}=\hat{i}-\hat{j}+3\hat{k}$ such that P is equidistant from the lines AB and AC. If $\left|\overrightarrow{AP}\right|=\frac{\sqrt{5}}{2}$, then the area of the triangle ABP is :

(1) 2 (2) $\frac{3}{2}$

(3) $\frac{\sqrt{30}}{4}$ (4) $\frac{\sqrt{26}}{4}$

18. Let

$A=\left\{z\in\mathbb{C}:|z-2|\le 4\right\}$ and

$B=\left\{z\in\mathbb{C}:|z-2|+|z+2|=5\right\}$.

Then the max $\left\{|z_1-z_2|:z_1\in A \text{ and } z_2\in B\right\}$ is

(1) $\frac{15}{2}$ (2) 8

(3) $\frac{17}{2}$ (4) 9

19. Let $f(x)=\int\frac{dx}{x^{\left(\frac{2}{3}\right)}+2x^{\left(\frac{1}{2}\right)}}$ be such that $f(0)=-26+24\log_e(2)$. If $f(1)=a+b\log_e(3)$, where $a, b\in\mathbf{Z}$, then $a+b$ is equal to:

(1) –18 (2) –5

(3) –11 (4) –26

20. $\frac{6}{3^{26}}+\frac{10.1}{3^{25}}+\frac{10.2}{3^{24}}+\frac{10.2^2}{3^{23}}+...+\frac{10.2^{24}}{3}$ is equal to

(1) 2^{25} (2) 2^{26}

(3) 3^{25} (4) 3^{26}

SECTION-B

21. If $\sum_{r=1}^{25}\left(\frac{r}{r^4+r^2+1}\right)=\frac{p}{q}$, where p and q are positive integers such that gcd (p, q) = 1, then p + q is equal to _____.

22. Three persons enter in a lift at the ground floor. The lift will go upto 10th floor. The number of ways, in which the three persons can exit the lift at three different floors, if the lift does not stop at first, second and third floors, is equal to _____ .

23. Let f be a differentiable function satisfying

$f(x)=1-2x+\int_0^x e^{(x-t)}\ f(t)\,dt$, $x\in\mathbf{R}$ and let

$g(x)=\int_0^x (f(t)+2)^{15}\ (t-4)^6\ (t+12)^{17}\,dt$, $x\in\mathbf{R}$.

If p and q are respectively the points of local minima and local maxima of g, then the value of |p + q| is equal to _____ .

24. If the distance of the point P(43, α, β), β < 0, from the line $\vec{r}=4\hat{i}-\hat{k}+\mu(2\hat{i}+3\hat{k})$, $\mu\in\mathbf{R}$ along a line with direction ratios 3, –1, 0 is $13\sqrt{10}$, then $\alpha^2+\beta^2$ is equal to _____.

25. Let $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$ and B be two matrices such that $A^{100}=100B+I$. Then the sum of all the elements of B^{100} is _____.



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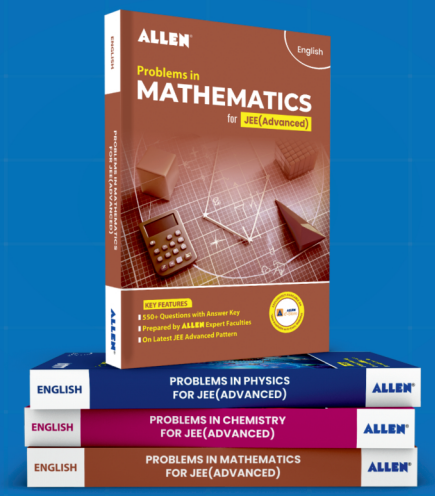


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